Vinayak Tyagi

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Education

Manipal University Jaipur Jaipur, India
B.Tech, Computer Science & Engineering | CGPA: 8.06 *2024 – 2028*

Experience

DevOps Intern Aug 2025 – Present
Software Development Centre, MUJ *Jaipur, India*

- Remediated **CVE-2025-55182** (CVSS 10.0, unauthenticated RCE in React Server Components) across **8 production services** within **24 hours** of public disclosure, and moved every Docker image to non-root execution in the same rollout.
- Rebuilt Dockerfiles and wrote incremental **GitHub Actions** build workflows across **45 repositories**, cutting image sizes by up to **82%** (2 GB → 358 MB) and **79%** (786 MB → 164 MB) and reducing registry storage and deploy times.
- Removed all public SSH exposure by moving Hostinger and bare-metal nodes onto a **Tailscale** zero-trust mesh; enforced per-service limits with **cgroups** so co-located containers cannot starve each other.
- Run deployment infrastructure for **3 production university platforms** (NPTEL certificate verification, elective management, IT ticketing) serving **2,160 CSE students**.

Projects

MIRAGE | *Internet-Facing SSH & Telnet Honeypot and Attacker-Measurement Platform* Live
Go • PostgreSQL • Python • SQL • Docker Compose • GitHub Actions • Cloudflare Tunnel Preprint & dataset
2-person team; I own the Go servers, shell, data pipeline, deployment and audit *Cited at SANSFIRE '26*

- Built the SSH server in **Go** on `x/crypto/ssh`: credential auth and a stateful fake shell (pipes, redirects, chaining, `$(...)`) over one filesystem with per-session randomized host identity; **320K+ real attacker sessions** recorded across two deployments.
- Added a **Telnet** sensor as an isolated service reusing the session, storage and shell code: a bounded, **fuzz-tested** RFC 854 negotiation state machine that fingerprints each client's options, terminal type and window size, plus a BusyBox persona for Mirai-style bots.
- **First-authored a preprint** auditing our 154,806-session corpus. Testing per-IP session counts for *divisibility*, not equality, exposed one scripted campaign behind **37.4%** of all sessions: 50 IPs across 4+ hosting providers, each an exact multiple of a shared 761-entry wordlist. A 257-IP negative control confirmed the test's specificity.
- Traced a silent capture defect (auth accepting every credential for 25 days, 73,243 attempts) to its exact boundary, then shipped live **data-validity checks** (accept-rate drift, cardinality collapse, campaign decomposition, downtime-vs-silence heartbeat).
- Hardened it against live adversarial traffic: fixed a race corrupting session state across concurrent SSH channels, bounded connections, channels and the DB pool against flood-triggered OOM, added per-connection panic recovery, and closed an unauthenticated `/metrics` endpoint and a case/path bypass (`WGET`, `/usr/bin/wget`) in the LLM-shell egress denylist.
- Built the **PostgreSQL** session pipeline and a cursor-paginated, streaming REST API; publish a weekly CC BY 4.0 dataset (IP anonymization, secret scrubbing) via **GitHub Actions**, with verified daily backups replicated offsite.

MUJ Placement Kite | *Placement Cell Platform* | Go • PostgreSQL • Redis • Cloudinary

- Built the student module end-to-end (profiles, draft-request approval workflow, RBAC, rate limiting, Cloudinary document uploads) and an idempotent bulk-upload pipeline that generically parses UG/PG Excel templates.
- Designed audit logging across **12 write paths** and placement-cycle hiring-workflow tracking (round pass/fail, roster + Excel export); built an SMTP worker pool over **Redis Streams** for async mail delivery.

NovaForge | *Multi-Factor Approval & Audit Platform* | TypeScript • Node.js • Prisma • React *Tally CodeBrewers — Finalist*

- Built **WebAuthn** passkey login (TOTP fallback, scrypt-hashed recovery codes) and an M-of-N / weighted-quorum approval engine with fail-closed fallback, recorded in a hash-chained, tamper-evident audit log of WebAuthn-signed votes.

Open Source

Honeynet Project | *EventHorizon* | github.com/honeynet/EventHorizon

- Added attacker-input capture to the **Telnet tarpit** in **C** (sanitized to printable ASCII, clean disconnect handling), exported as a per-IP counter by the **Go** Prometheus exporter; merged (PR #15); IAC-randomization and hardening PRs open.

Technical Skills

Languages: Go, Python, SQL, Bash, C, TypeScript

Networking & Security: TCP/IP, SSH & Telnet, protocol fuzzing, honeypots, container hardening, zero-trust networking

Infrastructure: Linux, Docker Compose, GitHub Actions, cgroups, Tailscale, Cloudflare Tunnel, Caddy, Prometheus, Grafana

Databases & Frameworks: PostgreSQL, Redis Streams, MongoDB, SQLite, Supabase, FastAPI, Next.js